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Important Equations In Series Resonant Circuit

$$V = V_m \sin \omega t$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$\text{Resonant frequency, } f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{Current, } I_0 = \frac{V}{R}$$

$$\text{Voltage across Capacitor, } V_{C0} = I_0 \times X_{C0} = I_0 / \omega_0 C$$

$$\text{Voltage across Inductor, } V_{L0} = I_0 \times X_{L0} = I_0 \omega_0 L$$

$$\text{Power factor, } \cos \phi = 1$$

$$\text{Power, } P_0 = I_0^2 R$$

$$\text{Quality Factor, } Q_0 = \frac{1}{R} \times \sqrt{\frac{L}{C}} \quad \text{or} \quad \frac{f_0}{\text{B.W}}$$

$$\text{Bandwidth, } B.W = \omega_2 - \omega_1 = R/L \quad (\text{rad/sec})$$

$$B.W = f_2 - f_1 = \frac{R}{2\pi L} \quad (\text{Hz})$$

~~Formula~~

$$\text{Lower cut-off frequency, } f_1 = f_0 - \frac{B.W}{2} = f_0 - \frac{R}{4\pi L}$$

$$\text{Upper cut-off frequency, } f_2 = f_0 + \frac{B.W}{2} = f_0 + \frac{R}{4\pi L}$$

$$\text{Lower cut-off frequency, } \omega_1 = \omega_0 - \frac{B.W}{2} = \omega_0 - \frac{R}{2L}$$

$$\text{Upper cut-off frequency, } \omega_2 = \omega_0 + \frac{B.W}{2} = \omega_0 + \frac{R}{2L}$$

Example 5.1 A series RLC circuit has the following parameter values: $R = 10 \, \Omega$, $L = 0.01 \, \text{H}$, $C = 100 \, \mu\text{F}$. Compute the resonant frequency, bandwidth, and lower and upper frequencies of the bandwidth.

Solution

$$R = 10 \, \Omega, \quad L = 0.01 \, \text{H}, \quad C = 100 \, \mu\text{F}$$

(a) Resonant frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.01 \times 100 \times 10^{-6}}} = 159.15 \, \text{Hz}$$

(b) Bandwidth

$$BW = \frac{R}{2\pi L} = \frac{10}{2\pi \times 0.01} = 159.15 \, \text{Hz}$$

(c) Lower frequency of bandwidth

$$f_1 = f_0 - \frac{BW}{2} = 159.15 - \frac{159.15}{2} = 79.58 \, \text{Hz}$$

(d) Upper frequency of bandwidth

$$f_2 = f_0 + \frac{BW}{2} = 159.15 + \frac{159.15}{2} = 238.73 \, \text{Hz}$$

Example 5.2 An RLC series circuit with a resistance of $10\ \Omega$, inductance of 0.2 H and a capacitance of $40\ \mu\text{F}$ is supplied with a 100 V supply at variable frequency. Find the following w.r.t. the series resonance circuit:

- frequency of which resonance takes place
- current
- power
- power factor
- voltage across RLC at that time
- quality factor
- half-power points
- resonance and phasor diagrams

Solution

$$R = 10\ \Omega, \quad L = 0.2\text{ H}, \quad C = 40\ \mu\text{F}, \quad V = 100\text{ V}$$

- (a) Resonant frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.2 \times 40 \times 10^{-6}}} = 56.3\text{ Hz}$$

- (b) Current

$$I_0 = \frac{V}{R} = \frac{100}{10} = 10\text{ A}$$

- (c) Power

$$P_0 = I_0^2 R = (10)^2 \times 10 = 1000\text{ W}$$

- (d) Power factor

$$\text{pf} = 1$$

- (e) Voltage across R, L, C

$$V_{R_0} = R I_0 = 10 \times 10 = 100\text{ V}$$

$$V_{L_0} = X_{L_0} I_0 = 2\pi f_0 L I_0 = 2\pi \times 56.3 \times 0.2 \times 10 = 707.5\text{ V}$$

$$V_{C_0} = X_{C_0} I_0 = \frac{1}{2\pi f_0 C} I_0 = \frac{1}{2\pi \times 56.3 \times 40 \times 10^{-6}} \times 10 = 707.5\text{ V}$$

- (f) Quality factor

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{10} \sqrt{\frac{0.2}{40 \times 10^{-6}}} = 7.07$$

- (g) Half-power points

$$f_1 = f_0 - \frac{R}{4\pi L} = 56.3 - \frac{10}{4\pi \times 0.2} = 52.32\text{ Hz}$$

$$f_2 = f_0 + \frac{R}{4\pi L} = 56.3 + \frac{10}{4\pi \times 0.2} = 60.3\text{ Hz}$$

(h) Resonance and phasor diagram are shown in Fig. 5.6.

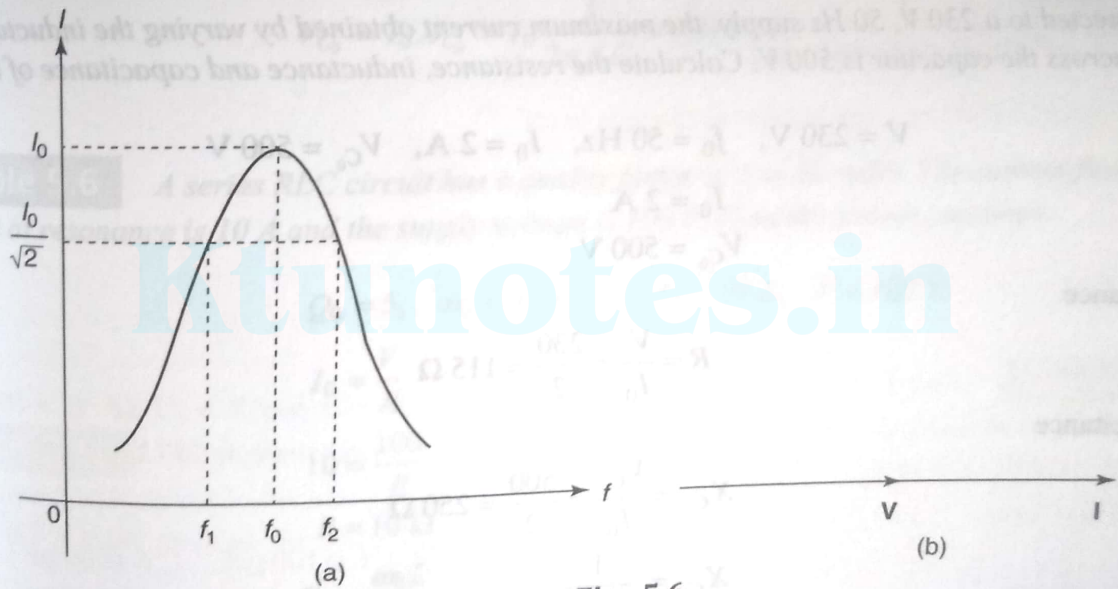


Fig. 5.6

Example 5.3

A series RLC circuit is connected to a 200 V ac supply. The current drawn by the circuit at resonance is 20 A. The voltage drop across the capacitor is 5000 V at series resonance. Calculate resistance and inductance if capacitance is 4 μF . Also, calculate the resonant frequency.

Solution

$$V = 200 \text{ V}, \quad I_0 = 20 \text{ A}, \quad V_{C_0} = 5000 \text{ V}, \quad C = 4 \mu\text{F}$$

(a) Resistance

$$R = \frac{V}{I_0} = \frac{200}{20} = 10 \Omega$$

(b) Resonant frequency

$$X_{C_0} = \frac{V_{C_0}}{I_0} = \frac{5000}{20} = 250 \Omega$$

$$X_{C_0} = \frac{1}{2\pi f_0 C}$$
$$250 = \frac{1}{2\pi \times f_0 \times 4 \times 10^{-6}}$$
$$f_0 = 159.15 \text{ Hz}$$

(c) Inductance
At resonance

$$X_{C_0} = X_{L_0} = 250 \Omega$$

$$X_{L_0} = 2\pi f_0 L$$

$$250 = 2\pi \times 159.15 \times L$$

$$L = 0.25 \text{ H}$$



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Example 5.4

A resistor and a capacitor are connected in series with a variable inductor. When the circuit is connected to a 230 V, 50 Hz supply, the maximum current obtained by varying the inductance is 2 A. The voltage across the capacitor is 500 V. Calculate the resistance, inductance and capacitance of the circuit.

Solution

$$V = 230 \text{ V}, \quad f_0 = 50 \text{ Hz}, \quad I_0 = 2 \text{ A}, \quad V_{C_0} = 500 \text{ V}$$

$$I_0 = 2 \text{ A}$$

$$V_{C_0} = 500 \text{ V}$$

(a) Resistance

$$R = \frac{V}{I_0} = \frac{230}{2} = 115 \, \Omega$$

(b) Capacitance

$$X_{C_0} = \frac{V_{C_0}}{I_0} = \frac{500}{2} = 250 \, \Omega$$

$$X_{C_0} = \frac{1}{2\pi f_0 C}$$

$$250 = \frac{1}{2\pi \times 50 \times C}$$

$$C = 12.73 \, \mu\text{F}$$

(c) Inductance
At resonance

$$X_{C_0} = X_{L_0} = 250 \, \Omega$$

$$X_{L_0} = 2\pi f_0 L$$

$$250 = 2\pi \times 50 \times L$$

$$L = 0.795 \text{ H}$$



(c) Voltage across capacitor

$$V_{C_0} = I_0 X_{C_0} = I_0 \frac{1}{2\pi f_0 C} = 100 \times \frac{1}{2\pi \times 50 \times 1013.2 \times 10^{-6}} = 314.16 \text{ V}$$

Example 5.6 A series RLC circuit has a quality factor of 5 at 50 rad/s. The current flowing through the circuit at resonance is 10 A and the supply voltage is 100 V. Find the circuit constants.

Solution

$$Q_0 = 5, \quad \omega_0 = 50 \text{ rad/s}, \quad I_0 = 10 \text{ A}, \quad V = 100 \text{ V}$$

$$I_0 = \frac{V}{R}$$

$$10 = \frac{100}{R}$$

$$R = 10 \Omega$$

$$Q_0 = \frac{\omega_0 L}{R}$$

$$5 = \frac{50 \times L}{10}$$

$$L = 1 \text{ H}$$

$$Q_0 = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$5 = \frac{1}{10} \sqrt{\frac{1}{C}}$$

$$C = 400 \mu\text{F}$$